I have the following code. Right now it is calculating for both on time and delay statistics and visuals. Can you adjust it for on time only:

```
{r}
library(dplyr)
library(tidyr)


# Step 1: Summarize totals
summary_df <- df %>%
  group_by(Airline, Status) %>%
  summarise(Total = sum(Count), .groups = "drop") %>%
  group_by(Airline) %>%
  mutate(Percentage = round((Total / sum(Total)) * 100, 2))
%>%
  ungroup()
summary_df

# Step 2: Wide format
wide_df <- summary_df %>%
  pivot_wider(
    names_from = Status,
    values_from = c(Total, Percentage),
    names_glue = "{Status}_{.value}"
  )
wide_df
```

```
# Step 3: Rename columns
wide_df <- wide_df %>%
  rename(
    Airline = Airline,
    `On Time Flights` = `on time_Total`,
    `Delayed Flights` = `delayed_Total`,
    `On Time (%)` = `on time_Percentage`,
    `Delayed (%)` = `delayed_Percentage`
  )

# Step 4: Display table
wide_df %>%
  gt() %>%
  tab_header(
    title = "Airline Performance Summary",
    subtitle = "On-Time vs Delayed Flights"
  ) %>%
  cols_align(
    align = "center",
    -Airline
  ) %>%
  fmt_number(
    columns = c(`On Time Flights`, `Delayed Flights`),
    decimals = 0
  ) %>%
  fmt_number(
    columns = c(`On Time (%)`, `Delayed (%)`),
    decimals = 2
  ) %>%
```

```
 text_transform(
   locations = cells_body(columns = c( On Time (%) ,
Delayed (%) )),
   fn = function(x) paste0(x, "%")
 )
```

The dataset looks like this:

Airline	Status	City	Count
ALASKA	on time	Los Angeles	479
ALASKA	on time	Phoenix	221
ALASKA	on time	San Diego	212
ALASKA	on time	San Francisco	503
ALASKA	on time	Seattle	1841
ALASKA	delayed	Los Angeles	62
ALASKA	delayed	Phoenix	12
ALASKA	delayed	San Diego	20
ALASKA	delayed	San Francisco	102
ALASKA	delayed	Seattle	305
AM WEST	on time	Los Angeles	694
AM WEST	on time	Phoenix	4840
AM WEST	on time	San Diego	383
AM WEST	on time	San Francisco	320
AM WEST	on time	Seattle	201
AM WEST	delayed	Los Angeles	117
AM WEST	delayed	Phoenix	415
AM WEST	delayed	San Diego	65
AM WEST	delayed	San Francisco	129
AM WEST	delayed	Seattle	61

Got it! You want to focus **only on on-time flights**, both in the summary statistics and the table/visuals, ignoring delayed flights entirely. We can simplify your code by filtering `Status == "on time"` early in the pipeline and dropping any columns related to delayed flights. Here's the adjusted code:

```
library(dplyr)
library(tidyr)
library(gt)

# Step 1: Filter for on-time flights and summarize totals
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    `On Time Flights` = sum(Count),
    .groups = "drop"
  ) %>%
  mutate(`On Time (%)` = round((`On Time Flights` / sum(`On Time Flights`)) * 10

# Step 2: Display table
on_time_df %>%
  gt() %>%
  tab_header(
    title = "Airline Performance Summary",
    subtitle = "On-Time Flights Only"
  ) %>%
  cols_align(
    align = "center",
    -Airline
  ) %>%
  fmt_number(
    columns = c(`On Time Flights`),
    decimals = 0
```

```r
    columns = c(`On Time (%)`),
    decimals = 2
  ) %>%
  text_transform(
    locations = cells_body(columns = c(`On Time (%)`)),
    fn = function(x) paste0(x, "%")
  )
```

✅ **Changes made:**

1. Filtered dataset to `Status == "on time"` at the start.
2. Removed any calculation or pivoting for delayed flights.
3. Table now only shows **On Time Flights** and **On Time (%)**.

If you want, I can also **adjust your visualizations** (bar charts, percentages, etc.) so they only show on-time performance. This will make the charts simpler and match your table.

Do you want me to do that too?

Do the same for the following barchart:
{r}
library(scales)

# Convert Status to factor with correct order
df_plot <- df_summary %>%
  mutate(Status = factor(Status, levels = c("on time",

```
"delayed"))) %>%
  pivot_longer(
    cols = c(Total, Percentage),
    names_to = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    # Round totals to integers
    Value = ifelse(Metric == "Total", round(Value), Value)
  )

# Create helper dataset for axis limits
df_limits <- df_plot %>%
  group_by(Metric) %>%
  summarise(
    Value = ifelse(
      Metric == "Percentage",
      100,              # force 0-100 for percentage
      max(Value) * 1.1   # add headroom for totals
    ),
    .groups = "drop"
  )

# Plot
ggplot(df_plot, aes(x = Airline, y = Value, fill = Status)) +
  geom_col(position = position_dodge(width = 0.8)) +

  # Add labels
  geom_text(
    aes(
```

```
    label = ifelse(
      Metric == "Total",
      comma(as.integer(Value)),
      paste0(round(Value, 2), "%")
    )
  ),
  position = position_dodge(width = 0.8),
  vjust = -0.3,
  size = 4
) +

# Axis limits per facet
geom_blank(
  data = df_limits,
  aes(y = Value),
  inherit.aes = FALSE
) +

facet_wrap(
  ~ Metric,
  scales = "free_y",
  labeller = labeller(
    Metric = c(
      "Total" = "Total Flights",
      "Percentage" = "Percentage (%)"
    )
  )
) +

# Custom colors
```

```
  scale_fill_manual(
    values = c(
      "on time" = "#87CEEB",   # blue
      "delayed" = "#FFA500"    # red
    ),
    labels = c("On Time", "Delayed")
  ) +

  labs(
    title = "Airline Performance Comparison",
    subtitle = "On-Time vs Delayed Flights (Total &
Percentage)",
    x = "Airline",
    y = NULL,
    fill = "Flight Status"
  ) +

  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

only keep on time

Perfect! To modify your bar chart for **on-time flights only**, we just need to:

1. Filter for `Status == "on time"` .

2. Remove any references to "delayed" in colors and labels.
3. Simplify the legend (or remove it if there's only one status).

Here's the adjusted code:

```
library(scales)
library(ggplot2)
library(dplyr)
library(tidyr)

# Filter for on-time flights only
df_plot <- df_summary %>%
  filter(Status == "on time") %>%
  pivot_longer(
    cols = c(Total, Percentage),
    names_to = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    # Round totals to integers
    Value = ifelse(Metric == "Total", round(Value), Value)
  )

# Create helper dataset for axis limits
df_limits <- df_plot %>%
  group_by(Metric) %>%
  summarise(
    Value = ifelse(
      Metric == "Percentage",
      100,             # 0-100 for percentages
      max(Value) * 1.1 # add headroom for totals
    ),
    .groups = "drop"
  )
```

```r
# Plot only on-time flights
ggplot(df_plot, aes(x = Airline, y = Value, fill = Status)) +
  geom_col(width = 0.6) +

  # Add labels
  geom_text(
    aes(
      label = ifelse(
        Metric == "Total",
        comma(as.integer(Value)),
        paste0(round(Value, 2), "%")
      )
    ),
    vjust = -0.3,
    size = 4
  ) +

  # Axis limits
  geom_blank(
    data = df_limits,
    aes(y = Value),
    inherit.aes = FALSE
  ) +

  facet_wrap(
    ~ Metric,
    scales = "free_y",
    labeller = labeller(
      Metric = c(
        "Total" = "Total Flights",
        "Percentage" = "Percentage (%)"
      )
    )
  ) +

  # Only one fill color for on-time
```

R

```r
    labels = c("On Time")
  ) +

  labs(
    title = "Airline Performance Summary",
    subtitle = "On-Time Flights Only",
    x = "Airline",
    y = NULL,
    fill = "Flight Status"
  ) +

  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

✅ **Key changes:**

- `filter(Status == "on time")` to exclude delayed flights.
- Removed `position_dodge()` since only one status exists.
- Only blue color for "on time" in `scale_fill_manual`.
- Updated title/subtitle to reflect **On-Time Flights Only**.

This will give you a **clean bar chart with totals and percentages for on-time flights only**.

If you want, I can also **merge the totals and percentages into a single bar per airline** for an even cleaner visual. It looks much neater when you have just one metric per airline.

Do you want me to do that?

Change the legend to ALASKA and AM WEST instead of On time since there is no delay stats anymore

Got it! Since we only have **on-time flights**, the legend should reflect the **Airline names** instead of flight status. This means we'll map `fill` to `Airline` instead of `Status` and remove references to "on time" in the color scale. Here's the updated code:

```
library(scales)
library(ggplot2)
library(dplyr)
library(tidyr)

# Filter for on-time flights only
df_plot <- df_summary %>%
  filter(Status == "on time") %>%
  pivot_longer(
    cols = c(Total, Percentage),
    names_to = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    # Round totals to integers
    Value = ifelse(Metric == "Total", round(Value), Value)
  )

# Create helper dataset for axis limits
df_limits <- df_plot %>%
  group_by(Metric) %>%
```

```
  summarise(
    Value = ifelse(
      Metric == "Percentage",
      100,             # 0-100 for percentages
      max(Value) * 1.1 # add headroom for totals
    ),
    .groups = "drop"
  )

# Plot on-time flights with legend as Airlines
ggplot(df_plot, aes(x = Airline, y = Value, fill = Airline)) +
  geom_col(width = 0.6) +

  # Add labels
  geom_text(
    aes(
      label = ifelse(
        Metric == "Total",
        comma(as.integer(Value)),
        paste0(round(Value, 2), "%")
      )
    ),
    vjust = -0.3,
    size = 4
  ) +

  # Axis limits
  geom_blank(
    data = df_limits,
    aes(y = Value),
    inherit.aes = FALSE
  ) +

  facet_wrap(
    ~ Metric,
    scales = "free_y",
    labeller = labeller(
```

```
      Metric = c(
        "Total" = "Total Flights",
        "Percentage" = "Percentage (%)"
      )
    )
  ) +

  # Assign colors per airline
  scale_fill_manual(
    values = c(
```

R

```
    )
  ) +

  labs(
    title = "Airline Performance Summary",
    subtitle = "On-Time Flights Only",
    x = "Airline",
    y = NULL,
    fill = "Airline"
  ) +

  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

✅ **What changed:**

1. `fill = Airline` → now each airline has its own color.
2. `scale_fill_manual()` assigns colors to **ALASKA** and **AM WEST**.
3. Legend title updated to `"Airline"` and shows the two airlines instead of "On Time".

This will make the chart more intuitive since **we only have on-time stats**, and the legend now identifies the airlines.

If you want, I can also **remove the facet entirely** and show **just one bar per airline with percentages on top**—it becomes a super clean single-chart view. Do you want me to do that too?

can you provide more optimized code that does the same thing

Absolutely! We can simplify and optimize your code by:

- Filtering for on-time flights right away.
- Pivoting only once and avoiding unnecessary `mutate` calls for rounding.
- Using `geom_col` with `aes(fill = Airline)` directly.
- Calculating axis limits more efficiently.

Here's a cleaner, optimized version:

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(scales)

# Prepare on-time data
df_plot <- df_summary %>%
```

```
  filter(Status == "on time") %>%
  pivot_longer(cols = c(Total, Percentage), names_to = "Metric", values_to = "Va
  mutate(
    Value = ifelse(Metric == "Total", round(Value), Value)
  )

# Calculate axis limits
axis_limits <- df_plot %>%
  group_by(Metric) %>%
  summarise(Value = ifelse(Metric == "Percentage", 100, max(Value) * 1.1), .grou

# Plot
ggplot(df_plot, aes(x = Airline, y = Value, fill = Airline)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = ifelse(Metric == "Total", comma(Value), paste0(round(Valu
            vjust = -0.3, size = 4) +
  geom_blank(data = axis_limits, aes(y = Value), inherit.aes = FALSE) +
  facet_wrap(~ Metric, scales = "free_y",
             labeller = labeller(Metric = c(Total = "Total Flights", Percentage =
  scale_fill_manual(values = c("ALASKA" = "#1f77b4", "AM WEST" = "#ff7f0e")) +
  labs(title = "Airline Performance Summary",
       subtitle = "On-Time Flights Only",
```

‹/› R

```
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

✅ Optimizations:

1. **Single** `pivot_longer` call.

2. **Rounded values in-place** instead of creating extra columns.
3. **Axis limits** calculated efficiently with one `summarise`.
4. **Legend now shows Airlines** without extra tweaks.
5. **Clean, readable structure**—less repetition, easier to maintain.

If you want, I can also create a **super compact version** that **removes facets** and shows **totals and percentages on the same bar chart**, which makes the visualization even cleaner for on-time only stats.

Do you want me to do that?

The percentage has changed now. Its not between on time and delay anymore, rather between two airlines. Modify the bar chart code based on the following table code:
{r}
```
library(dplyr)
library(tidyr)

# Step 1: Filter for on-time flights and summarize totals
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    Total Flights = sum(Count),
```

```
  .groups = "drop"
 ) %>%
 mutate( `Percentage (%)` = round(( `Total Flights` /
sum( `Total Flights` )) * 100, 2))

# Step 2: Display table
on_time_df %>%
 gt() %>%
 tab_header(
  title = "Airline Performance Summary",
  subtitle = "On-Time Flights Only"
 ) %>%
 cols_align(
  align = "center",
  -Airline
 ) %>%
 fmt_number(
  columns = c( `Total Flights` ),
  decimals = 0
 ) %>%
 fmt_number(
  columns = c( `Percentage (%)` ),
  decimals = 2
 ) %>%
 text_transform(
  locations = cells_body(columns = c( `Percentage (%)` )),
  fn = function(x) paste0(x, "%")
 )
```

Ah! I see exactly what you mean — the percentages should reflect **each airline's share of total on-time flights**, just like in your table, instead of being relative to each airline individually. Let's modify the bar chart to match that logic. We don't need facets anymore; we can plot **one bar per airline showing total flights**, and add **percentage labels on top**.

Here's the updated, clean code:

```
library(dplyr)
library(ggplot2)
library(scales)

# Prepare on-time summary data (matches your table)
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    `Total_Flights` = sum(Count),
    .groups = "drop"
  ) %>%
  mutate(`Percentage` = `Total_Flights` / sum(`Total_Flights`) * 100)

# Plot total flights with percentage labels
ggplot(on_time_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +

  # Add percentage labels on top
  geom_text(aes(label = paste0(round(Percentage, 2), "%")),
            vjust = -0.5, size = 5) +

  # Format y-axis numbers
  scale_y_continuous(labels = scales::comma) +

  # Assign custom colors to airlines
```

```
  scale_fill_manual(values = c("ALASKA" = "#1f77b4", "AM WEST" = "#ff7f0e")) +

  labs(
    title = "Airline Performance Summary",
    subtitle = "On-Time Flights Only",
```

‹/› R

```
    fill = "Airline"
  ) +

  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "top"
  )
```

✅ Key Changes:

1. Percentages are now **based on total on-time flights across all airlines** (`Total_Flights / sum(Total_Flights)`), exactly like your table.
2. Removed facets — the chart is now **one clean bar per airline**.
3. Percentage labels are displayed **on top of each bar**.
4. Custom colors for each airline, legend shows airline names.
5. Y-axis uses `comma()` for readable totals.

If you want, I can also **combine totals and percentages on the same bar using a secondary axis** to make it look even more polished and "dashboard-ready."

Do you want me to do that?

can you remove all unnecessary codes from this? I dont need scale anymore since I am only using one bar graph here. Make it beginner friendly and use minimum parameters

Absolutely! Since you just want a **simple, beginner-friendly bar chart** for on-time flights with percentages on top, we can strip it down to the essentials. Here's a minimal version:

```r
library(dplyr)
library(ggplot2)

# Prepare on-time summary data
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  mutate(Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2))

# Simple bar chart
ggplot(on_time_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col() +
```

‹/› R

```
    title = "On-Time Flights by Airline",
    x = "Airline",
    y = "Total Flights"
  )
```

✅ Why it's simple:

1. Only uses `geom_col` for the bars and `geom_text` for labels.
2. Percentages are shown on top of each bar.
3. Minimal `labs()` to add a title and axis labels.
4. No facets, no scales, no unnecessary parameters.

This is perfect for beginners — easy to read, modify, and extend later.

If you want, I can make an **even shorter 4–5 line version** that plots this without even using `mutate()` for percentage — super minimal for learning purposes.

Do you want me to do that?

show total flights as well with this viz

Perfect! We can show **both total flights (numeric) and percentage** on top of each bar in a beginner-friendly way. The simplest approach is to paste **Total and Percentage together** as labels. Here's a clean version:

```r
library(dplyr)
library(ggplot2)

# Prepare on-time summary data
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  mutate(Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2))

# Bar chart showing Total and Percentage
ggplot(on_time_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col() +
  geom_text(
    aes(label = paste0(Total_Flights, " (", Percentage, "%)")),
```

‹/› R

```r
  labs(
    title = "On-Time Flights by Airline",
    x = "Airline",
    y = "Total Flights"
  )
```

✅ What it does:

1. Each bar height = **total flights**.
2. Labels on top show: `Total Flights (Percentage%)`, e.g., `3260 (55%)`.
3. Minimal beginner-friendly code: no facets, no extra scales, just bars and labels.
4. `fill = Airline` gives each airline a different color automatically.

If you want, I can make an **even cleaner version** where **the bar itself represents percentage** and total flights are written as labels—super beginner-friendly and compact.

Do you want me to do that?

I want you to make it beautiful, for example it can be Total Flights = 6,438 and below that Percentage = 66.41%

Ah! Got it 😄 — you want **two lines of text on top of each bar**: one for total flights, one for percentage. We can do this using `\n` in the label. Here's a polished version:

```
library(dplyr)
library(ggplot2)
library(scales)

# Prepare on-time summary data
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  mutate(Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2))

# Beautiful bar chart with two-line labels
ggplot(on_time_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +
  geom_text(
```

```
    aes(label = paste0("Total Flights = ", comma(Total_Flights),
                       "\nPercentage = ", Percentage, "%")),
    vjust = -0.5, size = 4.5
  ) +
  labs(
```

R

```
    y = "Total Flights"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "none"
  )
```

✅ Improvements:

1. **Two-line labels** on top:

   Code

2. `comma()` makes large numbers readable.
3. Removed the legend since the airline is obvious on the x-axis.
4. Centered and bolded title for a cleaner look.
5. Slightly wider bars for better visibility.

This looks **dashboard-ready and beginner-friendly**, while keeping it simple.

If you want, I can also **add subtle gradient colors or custom colors per airline** to make it visually more attractive.

Do you want me to do that?

The values on top of the AM WEST got cut off by the title. Can you either make the y range higher or create space between title and the bar annotation

Absolutely! There are two simple ways to fix this:

1. **Increase the y-axis limits** so the bars and their labels have more space.
2. **Add extra space above the bars** using `expand` in `scale_y_continuous`.

Here's the updated beginner-friendly version with extra space above the bars so nothing gets cut off:

```
library(dplyr)
library(ggplot2)
library(scales)

# Prepare on-time summary data
on_time_df <- df %>%
  filter(Status == "on time") %>%
  group_by(Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  mutate(Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2))
```

```r
# Beautiful bar chart with space above labels
ggplot(on_time_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +
  geom_text(
    aes(label = paste0("Total Flights = ", comma(Total_Flights),
                       "\nPercentage = ", Percentage, "%")),
    vjust = -0.5, size = 4.5
  ) +
  labs(
    title = "On-Time Flights by Airline",
    x = "Airline",
```

R

```r
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +  # adds 15% space
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "none"
  )
```

✅ What changed:

- `scale_y_continuous(expand = expansion(mult = c(0, 0.15)))` adds **extra 15% space above the tallest bar**, so labels don't overlap with the title.
- Everything else stays the same — clean, beginner-friendly, beautiful two-line labels.

If you want, I can also **slightly move the title up** for extra breathing room, which makes it look even cleaner for presentations.

Do you want me to do that?

Now give me table code exactly as before. But this time Compare count and percentage arrival rates for two airlines across five cities.

Got it 👍 — you want the **same table style as before**, but now comparing **on-time arrival counts and percentages across cities for both airlines**.

So instead of grouping by Airline, we group by **Airline + City**, then calculate percentages **within each city** (so each city totals 100%). Then we reshape into a wide table.

Here is the clean version of your table code:

```
library(dplyr)
library(tidyr)
library(gt)

# Step 1: Filter on-time flights and summarize by Airline and City
city_df <- df %>%
  filter(Status == "on time") %>%
  group_by(City, Airline) %>%
  summarise(
    Total = sum(Count),
    .groups = "drop"
  ) %>%
  group_by(City) %>%
  mutate(
    Percentage = round((Total / sum(Total)) * 100, 2)
```

```
  ) %>%
  ungroup()

# Step 2: Convert to wide format
wide_df <- city_df %>%
  pivot_wider(
    names_from = Airline,
    values_from = c(Total, Percentage),
    names_glue = "{Airline}_{.value}"
  )

# Step 3: Rename columns (clean display)
wide_df <- wide_df %>%
  rename(
    City = City,
    `ALASKA Flights` = ALASKA_Total,
    `AM WEST Flights` = `AM WEST_Total`,
    `ALASKA (%)` = ALASKA_Percentage,
    `AM WEST (%)` = `AM WEST_Percentage`
  )

# Step 4: Display table
wide_df %>%
  gt() %>%
  tab_header(
    title = "Airline Arrival Performance by City",
    subtitle = "On-Time Arrival Counts and Percentages"
  ) %>%
  cols_align(
```

‹/› R

```
  fmt_number(
    columns = c(`ALASKA Flights`, `AM WEST Flights`),
    decimals = 0
  ) %>%
  fmt_number(
```

```
    columns = c(`ALASKA (%)`, `AM WEST (%)`),
    decimals = 2
  ) %>%
  text_transform(
    locations = cells_body(columns = c(`ALASKA (%)`, `AM WEST (%)`)),
    fn = function(x) paste0(x, "%")
  )
```

✅ What this table shows:

For each **city**:

- Total on-time flights for **ALASKA** and **AM WEST**
- Percentage share of on-time arrivals **within that city**

Example interpretation:

- In Los Angeles → ALASKA might have 41%, AM WEST 59%
- In Phoenix → AM WEST likely dominates

🔥 Why this is correct:

- Percentages are **city-based comparisons** (fair comparison across airlines)
- Matches your earlier table structure (wide format + gt styling)
- Keeps code **consistent, clean, and beginner-friendly**

If you want, I can also help you create a **matching bar chart for this city comparison** (which is actually very powerful for your assignment).

give me bar chart as before for this question too

Perfect 👍 — let's build a **bar chart matching your previous style**, but now for **city-level comparison between ALASKA and AM WEST**, using the same idea:

- Bar height = **Total Flights**
- Labels =

Code

- Percentages are calculated **within each city** (so ALASKA vs AM WEST per city)

✅ Clean, Beginner-Friendly Bar Chart Code

```
library(dplyr)
library(ggplot2)
library(scales)

# Step 1: Prepare data (same logic as table)
city_df <- df %>%
  filter(Status == "on time") %>%
```

```r
  group_by(City, Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  group_by(City) %>%
  mutate(
    Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2)
  ) %>%
  ungroup()

# Step 2: Bar chart (one plot per city)
ggplot(city_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +

  # Labels (two lines)
  geom_text(
    aes(label = paste0("Total Flights = ", comma(Total_Flights),
                       "\nPercentage = ", Percentage, "%")),
    vjust = -0.5, size = 3.8
  ) +

  # One panel per city
  facet_wrap(~ City) +

  # Add space above bars so labels don't get cut off
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +

  labs(
```

‹/› R

```r
    x = "Airline",
    y = "Total Flights"
  ) +

  theme_minimal(base_size = 13) +
  theme(
```

```
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "none",
    strip.text = element_text(face = "bold")
  )
```

✅ What this visualization shows:

- Each **panel = one city**
- Each panel has **two bars (ALASKA vs AM WEST)**
- Each bar shows:
  - Total flights (height)
  - Percentage share within that city (label)

🔥 Why this is great for your assignment:

- Direct comparison **within each city** (fair analysis)
- Matches your earlier **clean, minimal design**
- Easy to interpret:
  - Who dominates each city?
  - Where each airline performs better

💡 Optional improvement (if you want extra credit)

We can:

- Color ALASKA and AM WEST consistently across all plots
- Sort cities by performance

- Or highlight the winning airline per city

Just tell me 👍

Change this part and show percentage % and in bracket (total flights)

Got it 👍 — you want to **flip the label format** to:

Code

So instead of:

Code

You'll show:

Code

✅ Updated Label Code (only change is inside `geom_text()`)

Here is your updated bar chart with the new label format:

```
library(dplyr)
library(ggplot2)
library(scales)

# Prepare data
city_df <- df %>%
  filter(Status == "on time") %>%
  group_by(City, Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  group_by(City) %>%
  mutate(
    Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2)
  ) %>%
  ungroup()

# Bar chart
ggplot(city_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +

  # Updated labels (Percentage first, Total in brackets)
  geom_text(
    aes(label = paste0(Percentage, "%\n(", comma(Total_Flights), ")")),
    vjust = -0.5, size = 3.8
  ) +

  facet_wrap(~ City) +
```

```
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +

  labs(
    title = "On-Time Arrival Comparison by City",
```

R

```
    y = "Total Flights"
  ) +

  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "none",
    strip.text = element_text(face = "bold")
  )
```

✅ Result (Example Label):

Code

🔥 Why this looks better:

- Percentage is **highlighted first** (more important insight)
- Total flights are still visible but **less dominant**
- Cleaner and more professional for presentation

If you want, I can also **bold the percentage or make it larger than the total** for a more polished dashboard look 👍

change this part:
geom_text(
aes(label = paste0(comma(Total_Flights),
"\n", Percentage, "%")),
vjust = -0.5, size = 3.8
) +

Got it 👍 — you want to update **just the label format** so it shows:

Code

Here is the **modified** `geom_text()` **part only**:

```
geom_text(
  aes(label = paste0(Percentage, "%\n(", comma(Total_Flights), ")")),
```

R

```
  size = 3.8
) +
```

✅ Change made:

- Percentage shown first
- Total flights shown in parentheses on the next line
- Used `comma()` for readable numbers

This will display labels like:

Code

I am using this final code for barchart. But 3 barcharts goes on the first rown and then 2 cities barchart comes on the second row, It looks ugly. Can you sqeez all the bars in one row?

Yes 👍 — this happens because `facet_wrap()` automatically arranges panels into multiple rows. To **force all city charts into one single row**, you just need to specify the number of rows.

✅ Fix: Put all facets in one row

Just update your `facet_wrap()` like this:

‹/› R

🔧 Example (your full context)

```r
ggplot(city_df, aes(x = Airline, y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +

  geom_text(
    aes(label = paste0(Percentage, "%\n(", comma(Total_Flights), ")")),
    vjust = -0.5,
    size = 3.8
  ) +

  facet_wrap(~ City, nrow = 1) +   # 👈 all charts in one row

  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +

  labs(
    title = "On-Time Arrival Comparison by City",
    subtitle = "ALASKA vs AM WEST",
    x = "Airline",
    y = "Total Flights"
  ) +
```

‹/› R

```
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "none",
    strip.text = element_text(face = "bold")
  )
```

⚠️ Important tip

When you put everything in one row:

- The chart can get **wide and compressed**
- Text may look smaller

👉 If it looks too squeezed, you can:

- Increase figure width in R Markdown:

‹/› R

- Or reduce label size slightly:

‹/› R

🔥 Alternative (even cleaner)

If you want a **perfect grid (e.g., 5 cities)**, you can use:

R

This prevents small cities from looking flat.

If you want, I can make a **professional dashboard-style layout (all cities aligned with equal spacing and better readability)** 👍

Can you add airline names as legend and color but not in the graph? Also is it possible to create a stacked bar chart?

Great questions 👍 — let's handle both cleanly.

✅ 1. Add airline legend + colors (but remove labels from x-axis)

Right now, the airline names are on the x-axis. If you want them **only in the legend**, we:

- Use a **single x value** (so bars stack/group vertically)
- Keep `fill = Airline` → gives legend + color
- Remove x-axis labels

✅ Option A — Grouped bars (recommended)

```r
ggplot(city_df, aes(x = "", y = Total_Flights, fill = Airline)) +
  geom_col(position = "dodge", width = 0.6) +

  geom_text(
    aes(label = paste0(Percentage, "%\n(", comma(Total_Flights), ")")),
    position = position_dodge(width = 0.6),
    vjust = -0.5,
    size = 3.8
  ) +

  facet_wrap(~ City, nrow = 1) +

  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +

  labs(
    title = "On-Time Arrival Comparison by City",
    subtitle = "ALASKA vs AM WEST",
    x = NULL,
    y = "Total Flights",
    fill = "Airline"
  ) +
```

R

```r
    axis.text.x = element_blank(),   # remove x labels
    axis.ticks.x = element_blank(),
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

✅ Result:

- Bars still side-by-side
- No airline names on x-axis
- Legend shows **ALASKA / AM WEST**

✅ 2. Stacked Bar Chart (Yes — and it looks very clean)

A stacked chart is actually **excellent here** because:

- It shows **total flights per city**
- AND airline contribution visually

✅ Option B — Stacked bars

```
ggplot(city_df, aes(x = "", y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +

  geom_text(
    aes(label = paste0(Percentage, "%")),
    position = position_stack(vjust = 0.5),
    color = "white",
    size = 4
  ) +

  facet_wrap(~ City, nrow = 1) +

  labs(
    title = "On-Time Arrival Distribution by City",
```

```
    subtitle = "Percentage Share by Airline",
    x = NULL,
    y = "Total Flights",
    fill = "Airline"
  ) +
```

R

```
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

✅ Difference between the two:

Type	What it shows best
Grouped (side-by-side)	Compare airlines directly
Stacked	See share (%) within each city

🔥 Recommendation for your assignment

- Use **Grouped bar chart** → for comparison
- Use **Stacked bar chart** → for explanation

👉 Showing both gives you **extra credit level insight**

🚀 Pro Tip (very important)

For stacked charts, percentages are much more meaningful than totals.

If you want, I can make a **100% stacked bar chart (normalized)** — it looks very professional and is often expected in data analysis assignments 👍

turn off the grids behind

Nice finishing touch 👍 — removing the grid will make your chart look much cleaner.

Since you're using `theme_minimal()`, grids are turned on by default. You can **turn them off** using `theme()`.

✅ Remove all grid lines

Just add this to your plot:

```
theme(
```

‹/› R

🩹 Example (add to your existing code)

```
+ theme_minimal(base_size = 13) +
  theme(
    panel.grid = element_blank(),   # 👈 removes all grid lines
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
```

‹/› R

```
  )
```

🎯 Optional (clean but still structured)

If you want to keep **only horizontal lines** (better readability), use:

‹/› R

🔥 Professional look suggestion

For a polished dashboard style:

```
theme_minimal(base_size = 13) +
theme(
```

R

```
  legend.position = "top"
)
```

If you want, I can help you make the chart look like a **Tableau / Power BI style professional dashboard** 👍

give me the whole code in one shot

Absolutely 👍 — here is your **complete, clean, final code** for a **professional-looking stacked bar chart**:

- ✅ One row of cities
- ✅ Airline colors + legend (no x-axis labels)
- ✅ Percentage labels inside bars
- ✅ Clean look (no gridlines)
- ✅ Beginner-friendly

✅ Final Bar Chart Code

```r
library(dplyr)
library(ggplot2)
library(scales)

# Step 1: Prepare data
city_df <- df %>%
  filter(Status == "on time") %>%
  group_by(City, Airline) %>%
  summarise(
    Total_Flights = sum(Count),
    .groups = "drop"
  ) %>%
  group_by(City) %>%
  mutate(
    Percentage = round(Total_Flights / sum(Total_Flights) * 100, 2)
  ) %>%
  ungroup()

# Step 2: Stacked bar chart
ggplot(city_df, aes(x = "", y = Total_Flights, fill = Airline)) +
  geom_col(width = 0.6) +

  # Percentage labels inside bars
  geom_text(
    aes(label = paste0(Percentage, "%")),
    position = position_stack(vjust = 0.5),
    color = "white",
    size = 4
  ) +

  # One row of cities
  facet_wrap(~ City, nrow = 1) +

  labs(
```

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```
    y = "Total Flights",
    fill = "Airline"
```

R

```
  # Clean theme (no gridlines)
  theme_minimal(base_size = 13) +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line = element_line(color = "black"),
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "top",
    strip.text = element_text(face = "bold")
  )
```

🎯 What this gives you:

- Each **city = one panel**
- Each panel has **one stacked bar**
- Colors show **ALASKA vs AM WEST**
- Labels show **percentage share**
- Clean, **presentation-ready visualization**

🔥 If you want next-level polish

I can upgrade this to:

- **100% normalized bars (equal height for comparison)**
- **Custom airline colors**
- **Sorted cities by performance**
- **Highlight best airline per city**

Just tell me 👍

I fixed myself to the following code:

```
{r}
ggplot(city_df, aes(x = "", y = Total_Flights, fill = Airline)) +
  geom_col(position = "dodge", width = 0.6) +

  geom_text(
    aes(label = paste0(Percentage, "%\n(",
comma(Total_Flights), ")")),
    position = position_dodge(width = 0.6),
    vjust = -0.5,
    size = 2.5
  ) +

  facet_wrap(~ City, nrow = 1) +

  scale_y_continuous(expand = expansion(mult = c(0,
0.15))) +

  labs(
    title = "On-Time Arrival Comparison by City",
```

```
  x = NULL,
  y = "Total Flights",
  fill = "Airline"
) +

theme_minimal(base_size = 13) +
theme(
  panel.grid = element_blank(),
  axis.text.x = element_blank(),   # remove x labels
  axis.ticks.x = element_blank(),
  plot.title = element_text(face = "bold", hjust = 0.5),
  legend.position = "top",
  strip.text = element_text(face = "bold")
)
```

can you make the annotations bold?

Nice work — your chart is already very clean 👍

To make the **annotations (labels) bold**, you just need to add `fontface = "bold"` inside `geom_text()`.

✅ Updated `geom_text()` (bold labels)

```
geom_text(
  aes(label = paste0(Percentage, "%\n(", comma(Total_Flights), ")")),
```

```
  position = position_dodge(width = 0.6),
```

R

```
  fontface = "bold"    # 👈 makes text bold
) +
```

🎯 Optional (even better readability)

If you want the labels to stand out more, you can slightly increase size:

R

🔥 Bonus tip (professional look)

If labels are still hard to read against bars, you can add color:

R

or for darker bars:

R

If you want, I can help you **style labels like Tableau (bold percentage, smaller total)** for a very polished dashboard look 👍

Describe discrepancy between comparing two airlines' flight performances city-by-city and overall.Describe discrepancy between comparing two airlines' flight performances city-by-city and overall.

Here is the results:

Airline Performance Summary
On-Time Flights Only
Airline Total Flights Percentage (%)
ALASKA 3,256 33.59%
AM WEST 6,438 66.41%

here is the city results:
Airline Arrival Performance by City
On-Time Arrival Counts and Percentages
City ALASKA Flights AM WEST Flights ALASKA (%) AM WEST (%)
Los Angeles 479 694 40.84% 59.16%
Phoenix 221 4,840 4.37% 95.63%
San Diego 212 383 35.63% 64.37%
San Francisco 503 320 61.12% 38.88%
Seattle 1,841 201 90.16% 9.84%

Explain with example if there is any Simpson's Paradox

Great question — this is a classic situation where **aggregated results tell a different story than subgroup results**, and it's exactly where **Simpson's Paradox** can appear.

✅ 1. What your overall result says

From your summary:

Airline	Total Flights	Percentage
ALASKA	3,256	33.59%
AM WEST	6,438	66.41%

👉 **Conclusion (overall):**
AM WEST appears much better because it has **more on-time flights (66%)**.

✅ 2. What the city-by-city results say

City	Winner
Los Angeles	AM WEST
Phoenix	AM WEST

City	Winner
San Diego	AM WEST
San Francisco	ALASKA
Seattle	ALASKA

👉 **Conclusion (by city):**

- **AM WEST wins 3 cities**
- **ALASKA wins 2 cities (and very strongly in Seattle)**

BUT this is **not the full picture** — because we must consider **distribution of flights across cities**.

⚠️ 3. The discrepancy

🔴 **Overall:**

- AM WEST looks dominant (66%)

🟢 **Within cities:**

- Performance varies a lot
- ALASKA is **much better in some cities** (especially Seattle)

👉 So the ranking **changes depending on how you group the data**.

🔥 4. Why this happens (key insight)

The problem is **unequal distribution of flights across cities**.

Look at Phoenix:

- AM WEST: **4,840 flights**
- ALASKA: **221 flights**

👉 Phoenix alone dominates the dataset.

Since AM WEST performs well there:
➡️ It heavily boosts AM WEST's **overall percentage**

💡 5. This is Simpson's Paradox

Simpson's Paradox happens when:

> A trend appears in aggregated data but reverses or changes when the data is split into groups.

✅ In your data:

- **Overall:** AM WEST looks better
- **Within cities:** ALASKA is better in some important locations

👉 The conclusion depends on how you **group the data**

🧠 6. Clear example from your data

Example: Seattle

- ALASKA: **90.16%**
- AM WEST: **9.84%**

👉 ALASKA is **dramatically better**

BUT:

- AM WEST has very **few flights there (201)**
- ALASKA has **many flights (1,841)**

👉 This strong advantage does **not influence overall results enough**

Example: Phoenix

- AM WEST: **95.63%**
- ALASKA: **4.37%**

👉 AM WEST dominates

AND:

- It has **huge volume (4,840 flights)**

👉 This heavily skews the overall result

⚠️ 7. Why this is important

If you only look at overall data:

👉 You conclude **AM WEST is better**

But if you analyze by city:

👉 You see **performance depends on location**

🎯 8. Final explanation (ready to use in assignment)

You can write this:

> The overall comparison shows that AM WEST has a higher proportion of on-time flights (66.41%) compared to ALASKA (33.59%), suggesting that AM WEST performs better. However, when the data is analyzed by city, the results vary significantly. ALASKA outperforms AM WEST in cities such as San Francisco (61.12% vs 38.88%) and Seattle (90.16% vs 9.84%), while AM WEST performs better in Los Angeles, Phoenix, and San Diego.
>
> This discrepancy occurs because the distribution of flights across cities is uneven. AM WEST operates a large number of flights in Phoenix, where it has a very high on-time rate, which heavily influences the overall results. In contrast, ALASKA performs extremely well in Seattle but has fewer flights in cities where AM WEST dominates.
>
> This is an example of Simpson's Paradox, where aggregated data suggests one conclusion, but subgroup analysis reveals a different pattern. It highlights the importance of analyzing data at multiple levels before drawing conclusions.

🔥 Key takeaway

👉 **Overall averages can be misleading**

👉 Always check **group-level analysis**

If you want, I can help you turn this into a **perfect APA-style answer** for your assignment 👍

As final touch, I want the cities of this bargraph to appear at the bottom of the chart not top. Reedit the following code:

```
ggplot(city_df, aes(x = "", y = Total, fill = Airline)) +
  geom_col(position = "dodge", width = 0.6) +

  geom_text(
    aes(label = paste0(Percentage, "%\n(", comma(Total),
")")),
    position = position_dodge(width = 0.6),
    vjust = -0.5,
    size = 2.5,
    fontface = "bold"
  ) +

  facet_wrap(~ City, nrow = 1) +

  scale_y_continuous(expand = expansion(mult = c(0,
0.15))) +

  labs(
    title = "On-Time Arrival Comparison by City",
```

```
  x = NULL,
  y = "Total Flights",
  fill = "Airline"
) +

theme_minimal(base_size = 13) +
theme(
  panel.grid = element_blank(),
  axis.text.x = element_blank(),   # remove x labels
  axis.ticks.x = element_blank(),
  plot.title = element_text(face = "bold", hjust = 0.5),
  legend.position = "top",
  strip.text = element_text(face = "bold")
)
```

Good idea 👍 — moving city labels to the **bottom** makes the chart look much cleaner.

In `ggplot2`, facet labels (strip labels) are at the top by default. To move them to the **bottom**, you just need two changes:

✅ Key changes

1. Add `strip.position = "bottom"` to `facet_wrap()`
2. Tell the theme to place strips outside using:

‹/› R

✅ Updated Code

```r
ggplot(city_df, aes(x = "", y = Total, fill = Airline)) +
  geom_col(position = "dodge", width = 0.6) +

  geom_text(
    aes(label = paste0(Percentage, "%\n(", comma(Total), ")")),
    position = position_dodge(width = 0.6),
    vjust = -0.5,
    size = 2.5,
    fontface = "bold"
  ) +

  # Move city labels to bottom
  facet_wrap(~ City, nrow = 1, strip.position = "bottom") +

  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +

  labs(
    title = "On-Time Arrival Comparison by City",
    x = NULL,
    y = "Total Flights",
    fill = "Airline"
  ) +

  theme_minimal(base_size = 13) +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_blank(),
```

‹/› R

```r
    # Move strip labels outside and style them
```

```
    strip.placement = "outside",
    strip.text = element_text(face = "bold"),

    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "top"
  )
```

🎯 Result

- City names will now appear **below each panel**
- Chart looks more like a **dashboard / professional report**

🔥 Optional polish (recommended)

If you want the city labels to look like headers:

```
‹/› R
```

or add subtle background:

```
‹/› R
```

If you want, I can help you **turn this into a final presentation-ready figure (publication quality)** 👍

give me APA citation for this transcript

Here is an APA-style citation you can use for this ChatGPT conversation transcript:

APA Citation (7th Edition):

OpenAI. (2026, March 1). *ChatGPT conversation on airline performance analysis and Simpson's paradox* [Large language model]. ChatGPT. https://chat.openai.com/ ↗

In-text citation:

(OpenAI, 2026)

If you want, I can also format it specifically for your assignment (for example, including your course name like DATA 607 or adding a retrieval note).

Give me a conclusion in 3/4 lines for this overall analysis

The overall comparison suggests that AM WEST outperforms ALASKA in on-time arrivals; however, the city-by-city analysis reveals a more nuanced pattern where ALASKA performs better in several individual cities. This discrepancy indicates the presence of Simpson's Paradox, where aggregated data can mask underlying trends within subgroups. The imbalance in flight distribution across cities, particularly AM WEST's concentration in high-volume routes like Phoenix, skews the overall results. Therefore, performance should be evaluated at both the aggregate and city levels to avoid misleading conclusions.